Structural and Thermodynamic Validation of KRAS Allostery: A 100 ns Molecular Dynamics Comparison of Holo and Apo States

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https://github.com/Abdulwahab-12/pockets_kras_assessment

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Abstract

The oncogenic KRAS protein presents a significant therapeutic challenge due to its highly conserved orthosteric site. Targeting allosteric pockets, such as the Switch I/II cleft, represents a viable alternative. This study utilizes 100 ns Molecular Dynamics (MD) simulations to evaluate the structural integrity of KRAS (PDB: 4OBE) in both its Holo (GDP-bound) and Apo (ligand-free) states. Global equilibration metrics confirm system stability, while per-residue fluctuation (RMSF) analysis demonstrates a massive destabilization of the Switch I and Switch II loops upon GDP removal. Static and dynamic volumetric pocket analyses (`fpocket`/`mdpocket`) confirm that the Switch I/II allosteric cavity (Pocket 4) is highly druggable in the Holo state but undergoes severe structural collapse in the Apo state. This distinct phase-space separation demonstrates strict conformational selection, definitively validating Pocket 4 as a ligand-dependent allosteric target.

1 Introduction

KRAS is a critical GTPase that regulates cellular proliferation and is frequently mutated in human cancers [1]. Because the orthosteric binding site has a picomolar affinity for GTP/GDP, direct competitive inhibition has historically proven unfeasible [2]. Consequently, structure-based drug design has shifted toward identifying allosteric binding sites that can lock the protein in an inactive conformation [3]. One such putative target is the groove located between the Switch I and Switch II loops (Pocket 4) [3].

However, allosteric sites are often transient and heavily dependent on the global conformational state of the protein [4]. The objective of this study is to definitively validate the thermodynamic stability and druggability of Pocket 4. By comparing the Holo (GDP-bound) and Apo (ligand-free) states over a 100 ns trajectory, this report aims to prove that the orthosteric ligand acts as a necessary structural anchor, without which the allosteric pocket collapses.

2 Computational Methods

2.1 System Preparation and Molecular Dynamics

The Holo crystal structure of KRAS (PDB: 4OBE) was repaired to resolve missing heavy atoms. The Apo state was generated by stripping the HETATM (GDP/Mg^{2+}) records. Both systems were parameterized using the CHARMM36m force field [5], solvated in a TIP3P water box (dodecahedron, 1.0 nm edge distance), and neutralized with 0.15 M NaCl.

Following steepest descent energy minimization, the systems were equilibrated under NVT and NPT ensembles (310 K, 1 bar) using GROMACS [6]. A 100 ns unconstrained production MD trajectory was generated, centered, and stripped of periodic boundary conditions for analysis.

2.2 Allosteric Pocket Tracking

Initial static pocket detection was performed using `fpocket` [7]. To assess the dynamic viability of the validated allosteric target (Pocket 4), `mdpocket` [8] was utilized. The spatial coordinates of the Holo pocket were tracked across 100 evenly spaced trajectory snapshots (1 ns intervals) to quantitatively evaluate Volume, Solvent Accessible Surface Area (SASA), and Shape (alpha sphere density).

3 Results and Discussion

3.1 Global Stability and Local Flexibility

To confirm system equilibration and assess the structural impact of ligand removal, core trajectory metrics were evaluated over the 100 ns simulation (Figure 1). Backbone Root Mean Square Deviation (RMSD) analysis demonstrated that while both systems reached thermodynamic equilibrium, the Apo state deviated significantly more from the initial crystal structure, maintaining a consistently higher baseline.

This global structural loosening is quantitatively corroborated by the Radius of Gyration (Rg) and Solvent Accessible Surface Area (SASA) profiles. While neither system underwent complete unfolding, the Apo protein exhibited markedly higher Rg and SASA values throughout the entire trajectory compared to the tightly bound Holo state. This indicates that without the strong non-covalent interaction network provided by the GDP anchor, the entire protein matrix physically swells and relaxes, exposing substantially more internal surface area to the solvent.

Furthermore, local flexibility analyzed via Root Mean Square Fluctuation (RMSF) revealed that the Apo protein is not only globally expanded, but far more kinetically chaotic. The Apo state exhibited higher baseline fluctuations across the entire sequence, punctuated by massive destabilization spikes precisely at the Switch I (residues 30-40) and Switch II (residues 60-76) domains. This confirms that the removal of GDP releases these boundary loops to freely flail into the solvent space, directly compromising the topological integrity of the associated allosteric pocket.

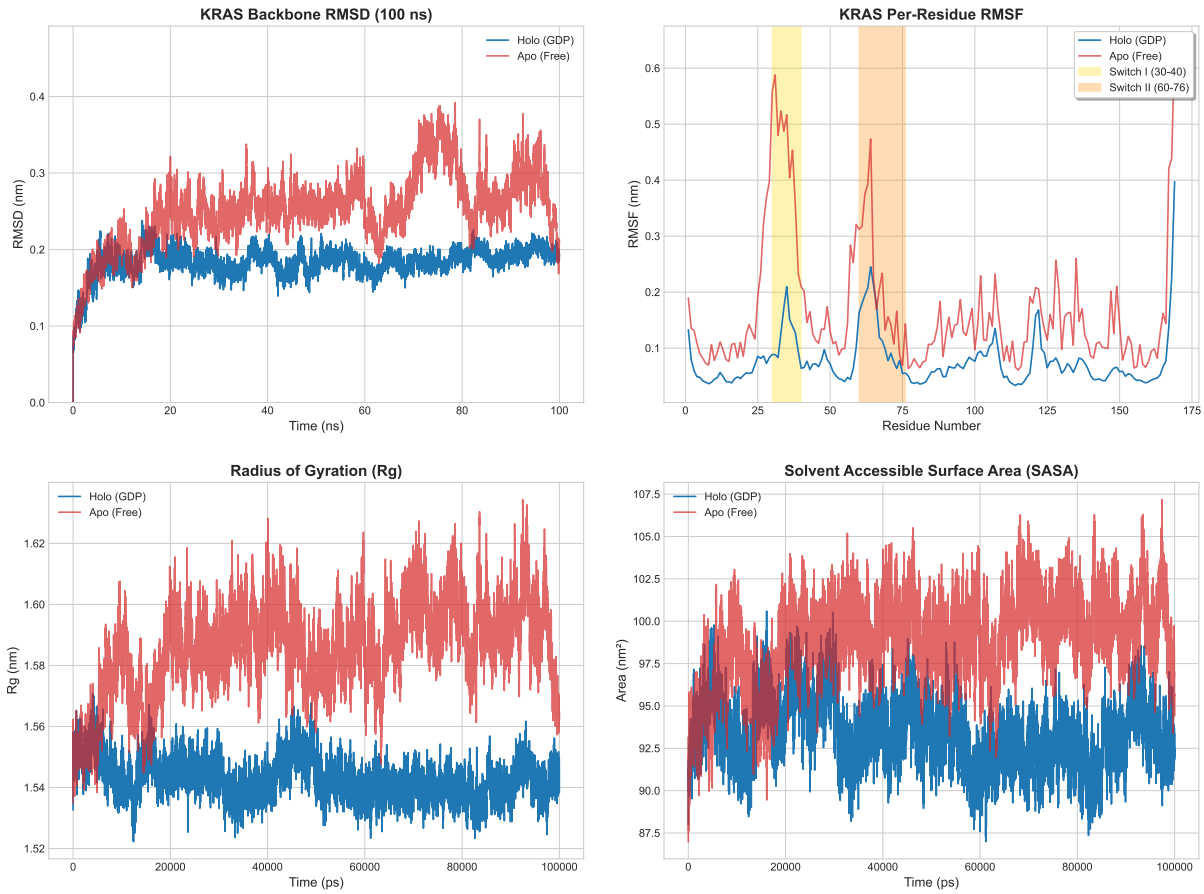


Figure 1: Core Trajectory Metrics. A 2x2 multi-panel comparison of the Holo (blue) and Apo (red) states. Stable RMSD, Rg, and SASA profiles confirm global protein folding is maintained, while the RMSF plot (top-right) highlights the massive localized destabilization of the Switch I and Switch II loops upon GDP removal.

3.2 Static Pocket Evaluation: Master FPocket Analysis

Prior to dynamic tracking, a comprehensive `fpocket` analysis was performed to evaluate every detectable cavity on the protein surface. This rigorous mapping allows us to separate transient, non-druggable surface artifacts from tangible, physiologically relevant binding sites.

As detailed in Table 1, initial crystal structures often present a multitude of topological grooves. The algorithm detected 8 cavities in the initial Holo state and 10 in the initial Apo state. However, the vast majority of these pockets (e.g., Pockets 1, 5, 6, 7, 8, 9, 10) are mere surface irregularities, presenting Druggability Scores near zero. The primary target of interest in this pre-equilibration phase is the Switch I/II Allosteric Site, which is initially identified by the algorithm as Pocket 3 prior to structural relaxation.

Table 1: Pre-Equilibration (Initial Crystal Structures) FPocket Analysis. Complete mapping of all detected cavities before thermodynamic relaxation. Bold rows highlight the primary allosteric target (Pocket 3) across both states prior to MD re-ranking.

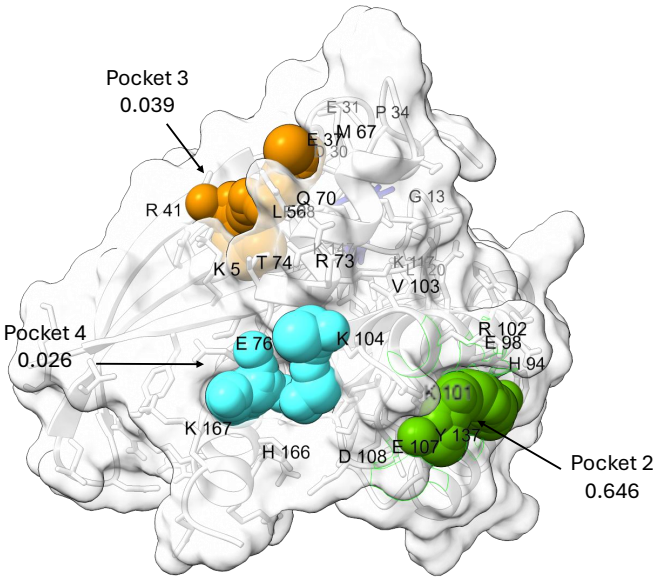
State & Pocket ID	Pocket Score	Druggability	Volume (Å^3)	Alpha Spheres	Total SASA	Apolar SASA
Initial Holo State (GDP-Bound)						
Pocket 1	0.313	0.002	369.65	44	116.57	39.85
Pocket 2	0.298	0.646	445.95	32	116.99	51.93
Pocket 3 (Switch I/II)	**0.245**	**0.039**	**390.65**	**30**	**121.89**	**59.17**
Pocket 4	0.240	0.026	468.63	30	128.96	66.42
Pocket 5	0.165	0.004	281.87	19	76.53	28.98
Pocket 6	0.145	0.006	424.77	24	115.01	50.72
Pocket 7	0.138	0.000	308.59	26	105.30	19.32
Pocket 8	0.099	0.000	213.83	17	73.78	6.04
Initial Apo State (Ligand-Free)						
Pocket 1	0.271	0.000	368.09	44	116.57	39.85
Pocket 2	0.269	0.026	456.68	32	116.99	51.93
Pocket 3 (Switch I/II)	**0.255**	**0.002**	**397.15**	**48**	**150.83**	**60.38**
Pocket 4	0.234	0.195	438.21	43	117.36	67.63
Pocket 5	0.217	0.002	398.99	30	121.89	59.17
Pocket 6	0.212	0.002	472.01	30	128.96	66.42
Pocket 7	0.148	0.000	288.05	19	76.53	28.98
Pocket 8	0.123	0.000	426.76	24	115.01	50.72
Pocket 9	0.114	0.000	310.09	26	105.30	19.32
Pocket 10	0.084	0.000	210.26	17	73.78	6.04

The critical filtering step occurs post-equilibration (Table 2). After physiological relaxation (Frame 0), the volumetric and topological shifts cause the `fpocket` algorithm to re-rank the cavities. In the Holo state, the Switch I/II target (now indexed as Pocket 4) successfully stabilizes into a highly viable conformation (Druggability: 0.387). Transient artifacts, such as the initial Holo Pocket 2, collapse to negligible druggability scores (0.008).

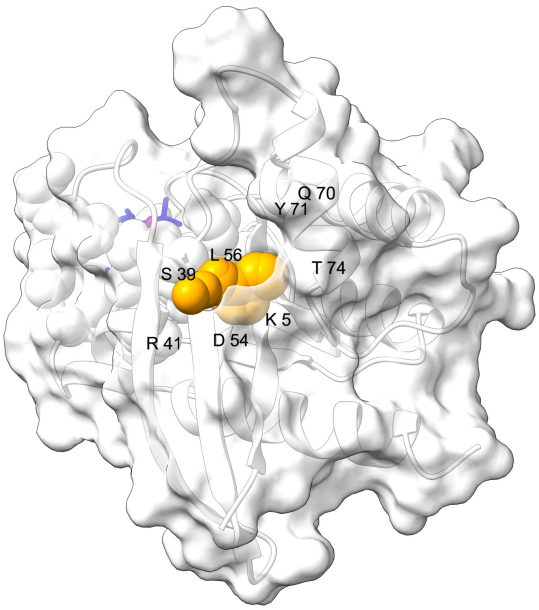
Conversely, the fully relaxed Apo structure demonstrates an immediate topological collapse of the allosteric target. Because the orthosteric GDP ligand is missing, the active site (Apo Pocket 2) expands into a massive, highly druggable crater (Score: 0.698, Volume: 877.90 Å^3). Without the stabilization of the ligand, the allosteric Switch I/II target (re-ranked as Pocket 4) is structurally crushed by the flailing loops, yielding a near-zero druggability score of 0.006.

3.3 Dynamic Allosteric Target Validation

To confirm that Pocket 4 does not permanently collapse during natural protein breathing, dynamic volumetric tracking was employed over the 100 ns trajectories. Figure 3 demonstrates



Initial Crystal Structure



Equilibrated Structure

Figure 2: Structural maturation of the allosteric binding site. Surface representations of the KRAS Holo state before and after MD equilibration. Top: The initial crystal structure. Bottom: Following thermodynamic relaxation.

Table 2: Post-Equilibration (MD Frame 0) FPocket Analysis. Complete mapping of all cavities after thermodynamic relaxation. Bold rows highlight the stabilization of the Holo allosteric target (re-ranked as Pocket 4) and the collapse of the corresponding Apo target, alongside the massive empty orthosteric site.

State & Pocket ID	Pocket Score	Druggability	Volume (Å^3)	Alpha Spheres	Total SASA	Apolar SASA
Equilibrated Holo State (GDP-Bound)						
Pocket 1	0.424	0.000	433.33	23	121.21	15.70
Pocket 2	0.421	0.008	141.16	25	49.59	20.53
Pocket 3	0.349	0.003	355.38	22	91.88	30.19
Pocket 4 (Switch I/II)	**0.315**	**0.387**	**169.89**	**20**	**65.56**	**27.78**
Pocket 5	0.140	0.000	232.60	15	94.30	42.27
Pocket 6	0.130	0.000	183.78	19	102.34	26.57
Equilibrated Apo State (Ligand-Free)						
Pocket 1	0.171	0.005	384.60	27	100.00	35.02
Pocket 2 (Empty Orthosteric)	**0.164**	**0.698**	**877.91**	**92**	**267.61**	**119.56**
Pocket 3	0.114	0.000	653.84	54	193.07	42.27
Pocket 4 (Switch I/II)	**0.113**	**0.006**	**297.35**	**24**	**99.52**	**41.06**
Pocket 5	0.101	0.006	70.55	15	10.91	9.82
Pocket 6	0.060	0.000	325.62	23	90.74	28.98
Pocket 7	-0.015	0.000	220.30	16	68.66	25.36
Pocket 8	-0.047	0.000	405.70	20	124.31	24.15
Pocket 9	-0.047	0.000	201.21	23	78.70	21.74
Pocket 10	-0.124	0.000	350.14	33	121.75	44.68

that in the presence of GDP, Pocket 4 is highly stable.

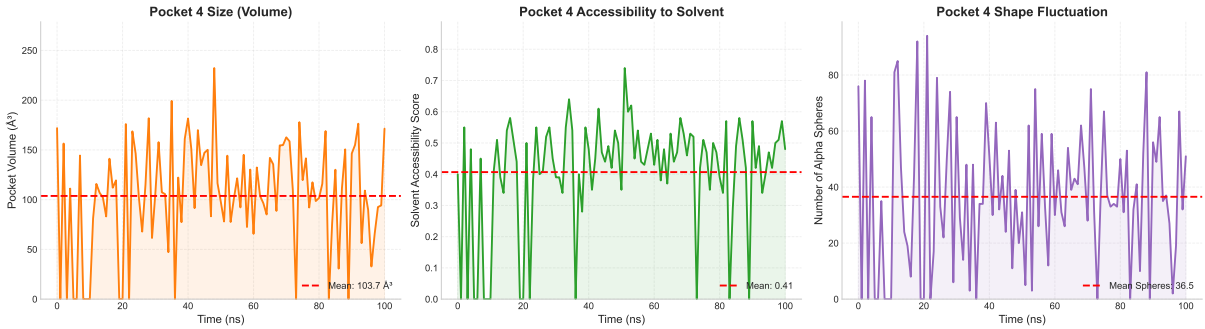


Figure 3: Holo Pocket Stability. Trajectory analysis confirming dynamic structural stability across Size, Accessibility, and Shape.

The ultimate proof of allosteric coupling is visually demonstrated by plotting the exact coordinates of Pocket 4 on the Apo trajectory. Because the boundary loops (S39, Y71) become highly flexible without GDP, they collapse inward, causing severe degradation of pocket volume (Figure 4).

This conformational selection is further mapped in a 3D phase space (Figure 5) and corresponding 2D density heatmap (Figure 6). The Apo trajectory rapidly shifts into an entirely different physical conformation, ensuring the two thermodynamic states remain strictly separated.

4 Conclusion

This comparative Molecular Dynamics study definitively confirms the allosteric coupling between the orthosteric GDP binding site and the distal Switch I/II pocket in KRAS. FPocket analysis revealed immediate collapse of the allosteric site upon ligand removal, while continuous tracking

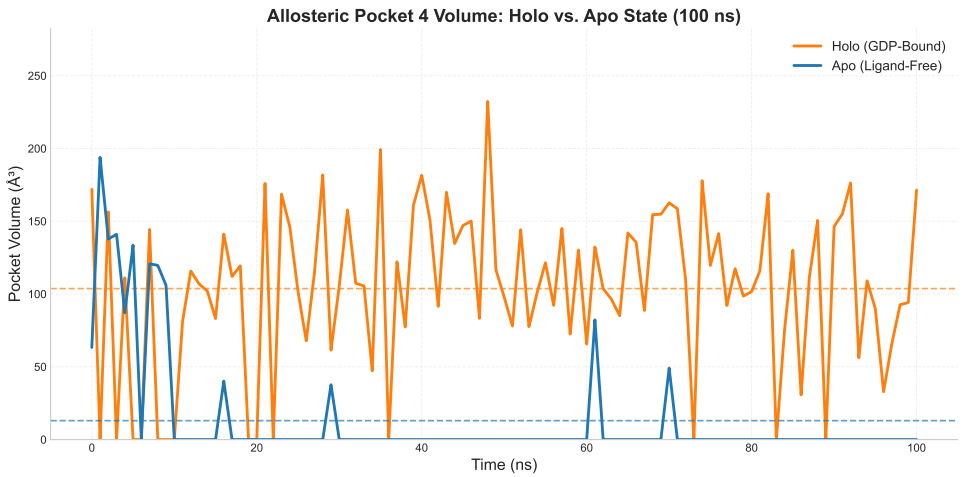


Figure 4: Allosteric Collapse. Volume overlay proving the destruction of the druggable cavity in the Apo state (blue).

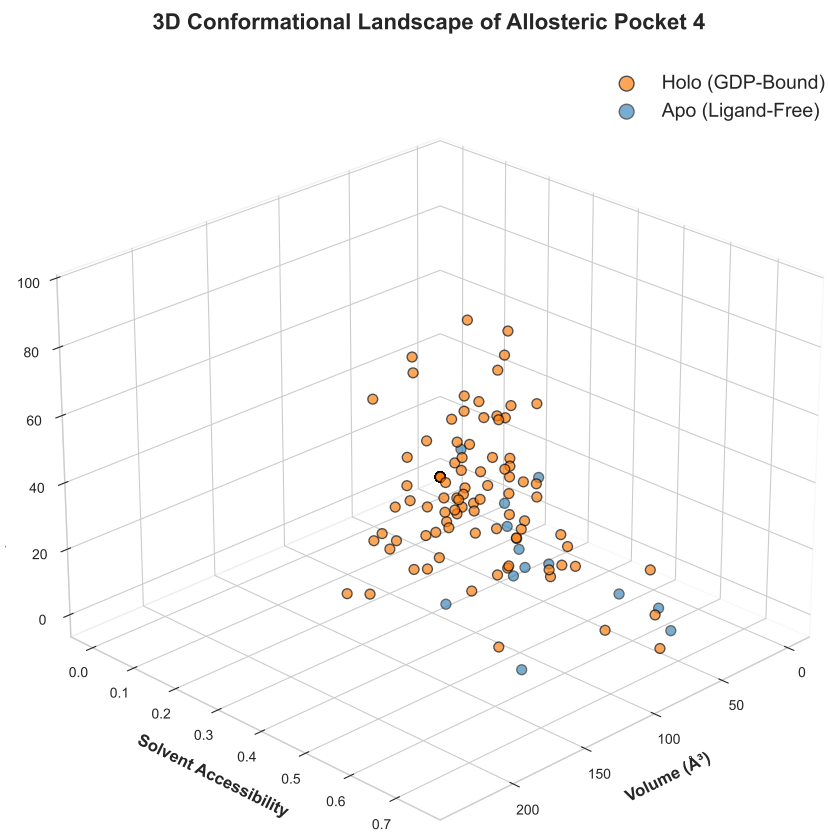


Figure 5: 3D Conformational Landscape. Mapping of Volume, Accessibility, and Shape demonstrating distinct phase-space separation.

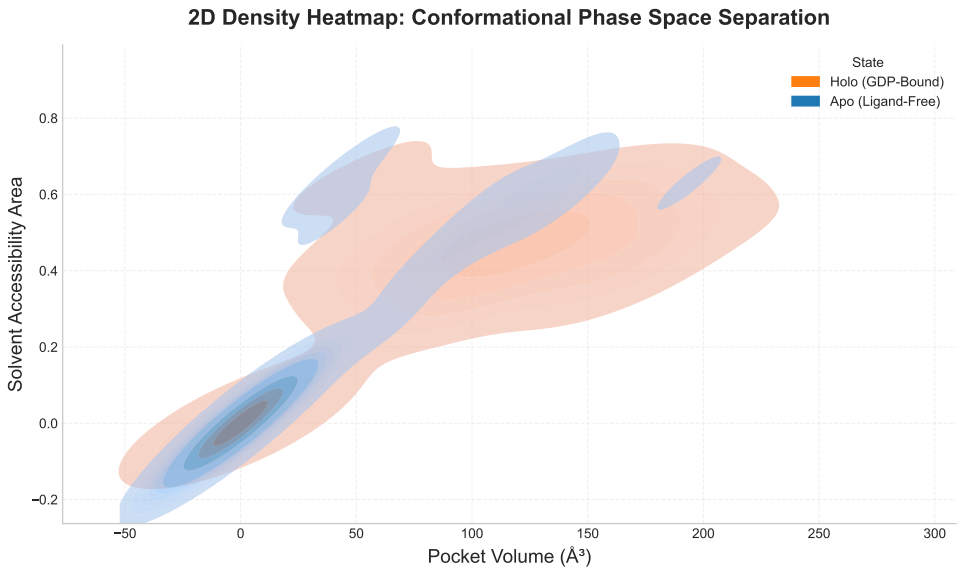


Figure 6: 2D Density Heatmap. A 2D Kernel Density Estimate (KDE) projecting the highest density clusters of the Holo (orange) and Apo (blue) conformational states, highlighting the severe loss of Volume and Accessibility in the ligand-free complex.

provided quantitative proof that the orthosteric ligand is required to restrict the flexibility of the Switch loops. The immediate and sustained structural collapse in the Apo state validates Pocket 4 as a strictly ligand-dependent allosteric target.

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